

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

December 2016

EC210.01 Analog Communication

Date: 07.12.2016, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** i. An AM signal is represented by $x(t) = (20+4\sin 500\pi t)\cos(2\pi 10^5 t)V$. The modulation index is [01]
(A) 20 (B) 4 (C) 0.2 (D) 10
- ii. An AM signal is represented by $x(t) = (20+4\sin 500\pi t)\cos(2\pi 10^5 t)V$. The total signal power is [01]
(A) 208 W (B) 204 W (C) 408 W (D) 416 W
- iii. An AM signal is represented by $x(t) = (20+4\sin 500\pi t)\cos(2\pi 10^5 t)V$. The total sideband power is [01]
(A) 4 W (B) 8 W (C) 16 W (D) 2 W
- iv. A 2 kW carrier is to be modulated to a 90% level. The total transmitted power would be [01]
(A) 3.62 kW (B) 2.81 kW (C) 1.4 kW (D) None of the above
- v. A 2 MHz carrier is amplitude modulated by a 500Hz modulating signal to a depth of 70%. If the unmodulated carrier power is 2 kW, the power of the modulated signal is [01]
(A) 2.23 kW (B) 2.36 kW (C) 1.18 kW (D) 1.26 kW
- vi. If the modulation index of an AM wave is changed from 0 to 1, the transmitted power [01]
(A) increases by 50% (B) increases by 75% (C) increases by 100%
(D) remains unaffected
- vii. In amplitude modulation, the modulation envelope has a peak value which is double the unmodulated carrier value. What is the value of the modulation index?
(A) 25% (B) 50% (C) 75% (D) 100%
- Q - 2 (a)** i. The antenna current of an AM transmitter is 8A when only carrier is sent. It increases to 8.93A when the carrier is modulated by a single sine wave. Find the percentage modulation. [03]
- ii. What are the advantages of DSB-SC and SSB-SC? [04]
- (b)** Draw and Explain block diagram of Amplitude Modulated transmitter with each block in detail. [07]

OR

- Q - 2 (a)** What is the relation between total power, unmodulated carrier power and modulation index? Derive its mathematical expressions. [07]

- (b) i. What is amplitude modulation and modulation index of AM? Draw the carrier, information and modulated wave with different modulation index ($m < 1$, $m = 1$, $m > 1$). [03]
 ii. What is the need for modulation? [02]
 iii. What are advantages of ring modulator? [02]
- Q - 3 (a) Compare Single sideband (SSB) modulation system with amplitude modulation (AM) system. [07]
 (b) Draw and explain the circuit diagram of ring or lattice modulator for DSBSC (double sideband suppressed carrier) generation. [07]

OR

- Q - 3 (a) Explain about the generation of SSB using filter method. [07]
 (b) Draw and explain block schematic of “phase shift” method for SSB generation. [07]

SECTION – II

- Q - 4 (a) i. A pre-emphasis circuit provides extra noise immunity by [01]
 (A) boosting the base frequencies (B) amplifying the higher audio frequencies
 (C) pre-amplifying the whole audio band (D) converting the phase modulation to FM
- ii. When the modulating frequency is doubled, the modulation index is halved and the modulating voltage remains constant. The modulation system is [01]
 (A) AM (B) PM (C) FM (D) none of the above [01]
- iii. An FM wave is 100% modulated by 10KHz signal. The bandwidth required is [01]
 (A) 170KHz (B) 20KHz (C) 7.5KHz (D) 75KHz
- iv. Power in an FM signal is dependent on [01]
 (A) the amplitude of the carrier (B) both amplitude and frequency of the carrier
 (C) amplitude and frequency of the carrier (D) none of the above [01]
- v. The modulating frequency in frequency modulation is increased from 10KHz to 20KHz the bandwidth [01]
 (A) decreases by 20KHz (B) increases by 20KHz
 (C) decreases by 10KHz (D) increases by 10KHz [01]
- vi. In commercial FM broadcasting, the maximum frequency deviation is [01]
 (A) 75KHz (B) 200KHz (C) 5KHz (D) 15KHz
- vii. Audio bandwidth for standard telephone is [01]
 (A) 30Hz to 3000Hz (B) 50Hz to 3400Hz (C) 300Hz to 3400Hz
 (D) 50Hz to 5000Hz
- Q - 5 (a) Compare frequency modulation with amplitude modulation. [07]
 (b) Explain typical deemphasis circuit with its application. [07]

OR

- Q - 5 (a) Differentiate directly modulated FM transmitter and indirectly modulated FM transmitter. [07]
 (b) In frequency modulating system the frequency deviation constant is 2 KHz/V. A sinusoidal modulating signal of amplitude 20V and frequency 4 KHz is applied. Calculate [07]
 i. Frequency deviation ii. Modulation index
 iii. Carrier swing

- Q - 6 (a)** Discuss the concept of frequency translation. [07]
- (b)** What is an AGC? Explain different types of AGC control used in a communication receiver. [07]

OR

- Q - 6 (a)** Explain Tuned Radio Frequency receiver in detail. [07]
- (b)** How do you adjust the budget of superheterodyne receiver? Explain with the help of an example? [07]
